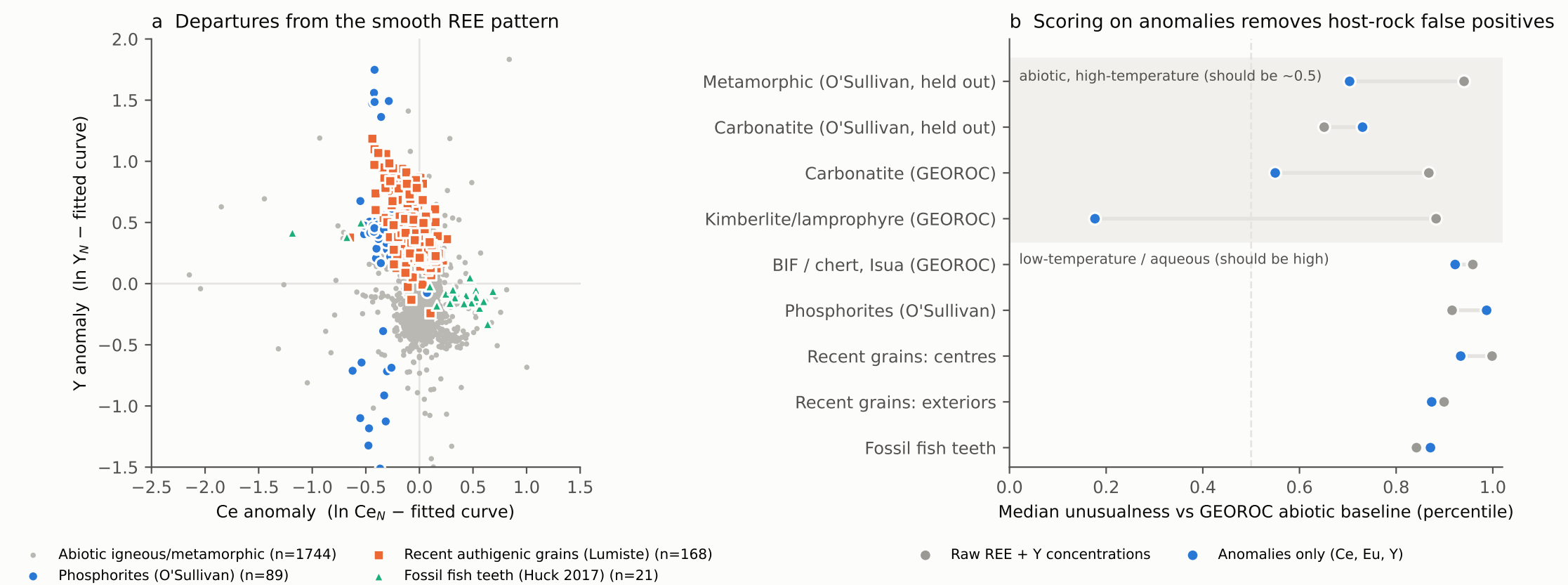

a Departures from the smooth REE pattern
Y anomaly (ln Y_N − fitted curve)
Ce anomaly (ln Ce_N − fitted curve)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
b Scoring on anomalies removes host-rock false positives
Metamorphic (O'Sullivan, held out)
Carbonatite (O'Sullivan, held out)
Carbonatite (GEOROC)
Kimberlite/lamprophyre (GEOROC)
BIF / chert, Isua (GEOROC)
Phosphorites (O'Sullivan)
Recent grains: centres
Recent grains: exteriors
Fossil fish teeth
abiotic, high-temperature (should be ~0.5)
low-temperature / aqueous (should be high)
0.0
0.2
0.4
0.6
0.8
1.0
Median unusualness vs GEOROC abiotic baseline (percentile)
Abiotic igneous/metamorphic (n=1744)
Phosphorites (O'Sullivan) (n=89)
Recent authigenic grains (Lumiste) (n=168)
Fossil fish teeth (Huck 2017) (n=21)
Raw REE + Y concentrations
Anomalies only (Ce, Eu, Y)